

PENRITH HIGH SCHOOL



MATHEMATICS EXTENSION 2 2012

HSC Trial

Assessor: Mr Ferguson

General Instructions:

- Reading time – 5 minutes
- Working time – **3 hours**
- Write using black or blue pen. Black pen is preferred
- Board-approved calculators may be used.
- A table of standard integrals is provided at the back of this paper.
- A multiple choice answer sheet is provided at the back of this paper.
- Show all necessary working in Questions 11 – 16.
- Work on this question paper will not be marked.

Section 1

Total marks – 100

SECTION 1 – Pages 2 – 5

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section.

SECTION 2 – Pages 6 – 12

90 marks

- Attempt Questions 11 – 16
- Allow about 2 hours 45 minutes for this section.

Section 2

Question	Mark
1	
2	
3	
4	
5	

Question	Mark
6	
7	
8	
9	
10	
Total	/10

Question	Mark
11	/15
12	/15
13	/15
14	/15
15	/15
16	/15

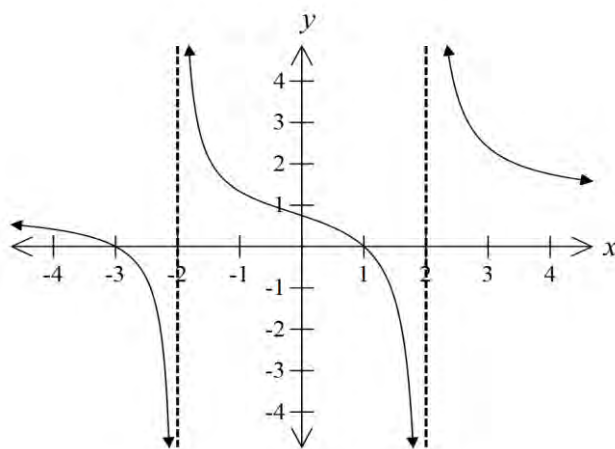
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This paper MUST NOT be removed from the examination room

Student Number:

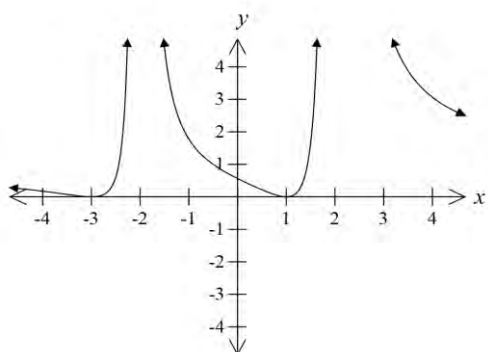
SECTION 1: Circle the correct answer on the multiple choice answer sheet

- 1 The diagram shows the graph of the function $y = f(x)$.

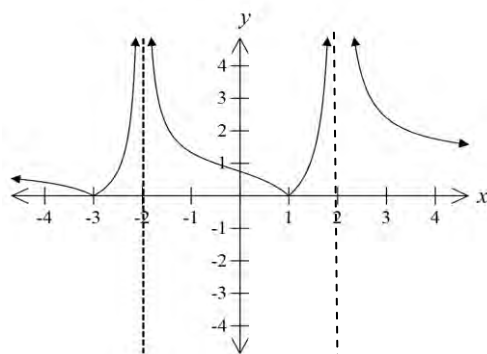


Which of the following is the graph of $y = |f(x)|$?

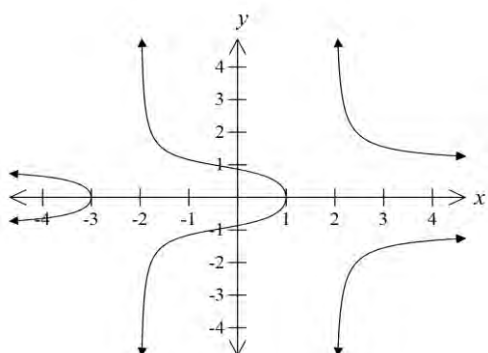
(A)



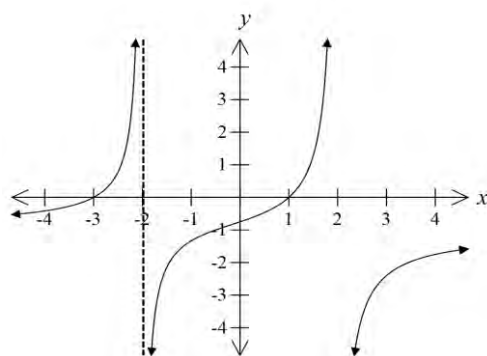
(B)



(C)



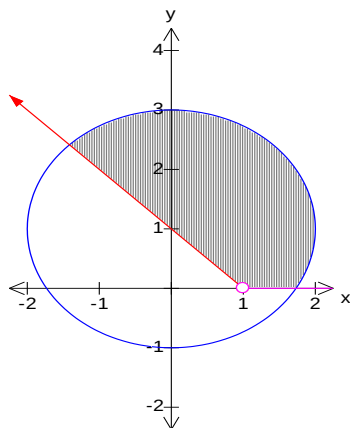
(D)



2 Let $z = 4 + i$. What is the value of $\overline{iz}$?

- (A) $-1 - 4i$
- (B) $-1 + 4i$
- (C) $1 - 4i$
- (D) $1 + 4i$

3 Consider the Argand diagram below.



Which inequality could define the shaded area?

- (A) $|z - i| \leq 2$ and $0 \leq \arg(z - 1) \leq \frac{3\pi}{4}$
- (B) $|z + i| \leq 2$ and $0 \leq \arg(z - 1) \leq \frac{3\pi}{4}$
- (C) $|z - i| \leq 2$ and $0 \leq \arg(z - 1) \leq \frac{\pi}{4}$
- (D) $|z + i| \leq 2$ and $0 \leq \arg(z - 1) \leq \frac{\pi}{4}$

4 Consider the hyperbola with the equation $\frac{x^2}{9} - \frac{y^2}{5} = 1$.

What are the coordinates of the vertex of the hyperbola?

- (A) $(\pm 3, 0)$
- (B) $(0, \pm 3)$
- (C) $(0, \pm 9)$
- (D) $(\pm 9, 0)$

5 The points $P(cp, \frac{c}{p})$ and $Q(cq, \frac{c}{q})$ lie on the same branch of the hyperbola $xy = c^2$ ($p \neq q$). The tangents at P and Q meet at the point T . What is the equation of the normal to the hyperbola at P ?

- (A) $p^2x - py + c - cp^4 = 0$
- (B) $p^3x - py + c - cp^4 = 0$
- (C) $x + p^2y - 2c = 0$
- (D) $x + p^2y - 2cp = 0$

6 What is the value of $\int \sec x dx$? Use the substitution $t = \tan \frac{x}{2}$.

- (A) $\ln |(t+1)(t-1)| + c$ (B) $\ln \left| \frac{1+t}{1-t} \right| + c$
(C) $\ln |(1+t)(1-t)| + c$ (D) $\ln \left| \frac{t+1}{t-1} \right| + c$

7 Let $I_n = \int_0^x \sin^n t dt$, where $0 \leq x \leq \frac{\pi}{2}$.

Which of the following is the correct expression for I_n ?

- (A) $I_n = \left(\frac{n-1}{n} \right) I_{n-2}$ with $n \geq 2$.
(B) $I_n = \left(\frac{n+1}{n} \right) I_{n-2}$ with $n \geq 2$.
(C) $I_n = n(n-1) I_{n-2}$ with $n \geq 2$.
(D) $I_n = n(n+1) I_{n-2}$ with $n \geq 2$.

8 The region enclosed by $y = x^3$, $y = 0$ and $x = 2$ is rotated around the y -axis to produce a solid. What is the volume of this solid?

- (A) $\frac{8\pi}{5}$ units³
(B) $\frac{32\pi}{5}$ units³
(C) $\frac{64\pi}{5}$ units³
(D) $\frac{16\pi}{5}$ units³

9 What is the angle at which a road must be banked so that a car may round a curve with a radius of 100 metres at 90 km/h without sliding? Assume that the road is smooth and gravity to be 9.8 ms^{-2} .

- (A) $83^\circ 10'$ (B) $32^\circ 32'$
(C) $83^\circ 6'$ (D) $32^\circ 53'$

- 10** The polynomial equation $x^3 + 4x^2 - 2x - 5 = 0$ has roots α , β and γ . Which of the following polynomial equations have roots α^2 , β^2 and γ^2 ?
- (A) $x^3 - 20x^2 - 44x - 25 = 0$
- (B) $x^3 - 20x^2 + 44x - 25 = 0$
- (C) $x^3 - 4x^2 + 5x - 1 = 0$
- (D) $x^3 + 4x^2 + 5x - 1 = 0$

SECTION 2

Question 11 (15 marks) (Use a new page to write your answers)

(a) Find (i) $\int \frac{t^2 - 1}{t^3} dt$. 4

(ii) $\int \frac{dx}{\sqrt{6 - x - x^2}}$

(b) Evaluate (i) $\int_0^1 \frac{x}{(x+1)(2x+1)} dx$ 3

(ii) $\int_0^{\frac{\pi}{4}} x \tan^2 x dx$ 3

(c) (i) If $I_n = \int_0^{\frac{\pi}{2}} x^n \cos x dx$, show that for $n > 1$, 3

$$I_n = \left(\frac{\pi}{2}\right)^n - n(n-1)I_{n-2}$$

(ii) Hence find the area of the finite region bounded by the curve 2

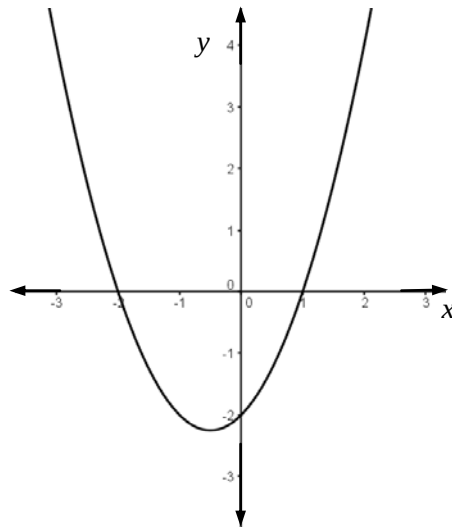
$y = x^4 \cos x$ and the x axis for $0 \leq x \leq \frac{\pi}{2}$.

Question 12 (15 marks) (Use a new page to write your answers)

- (a) Given that $z = \sqrt{2} - \sqrt{2}i$ and $w = -\sqrt{2}$, find, in the form $x + iy$:
- (i) wz^2 1
 - (ii) $\arg z$ 1
 - (iii) $\frac{z}{z+w}$ 2
 - (iv) $|z|$ 1
 - (v) z^{10} 2
- (b) Find the values of real numbers a and b such that $(a + ib)^2 = 5 - 12i$ 2
- (c) Draw Argand diagrams to represent the following regions 2
- (i) $1 \leq |z + 4 - 3i| \leq 3$
 - (ii) $\frac{\pi}{6} \leq \arg z \leq \frac{\pi}{3}$
- (d) (i) Show that $\frac{1 + \cos \theta + i \sin \theta}{1 - \cos \theta - i \sin \theta} = i \cot \frac{\theta}{2}$ 2
- (ii) Hence solve $\left(\frac{z-1}{z+1}\right)^8 = -1$ 2

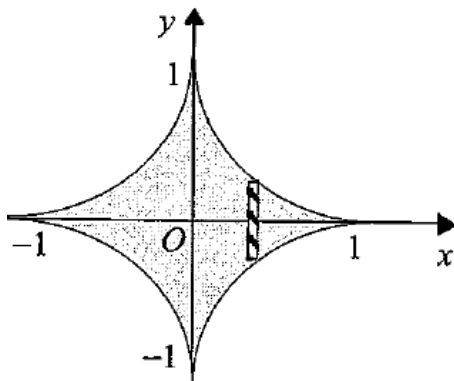
Question 13 (15 marks) (Use a new page to write your answers)

- (a) The diagram shows the graph of the function $f(x) = x^2 + x - 2$. On separate diagrams sketch the following graphs, showing clearly any intercepts on the coordinate axes and the equations of any asymptotes.

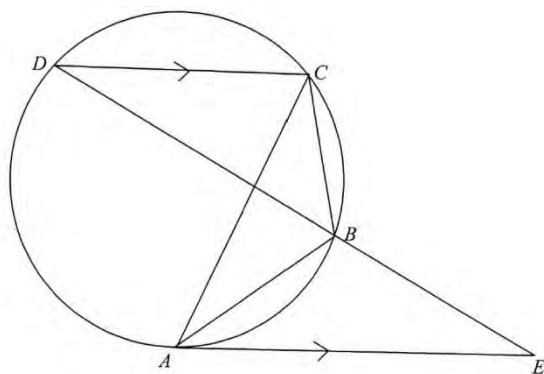


- | | | |
|-------|----------------------|---|
| (i) | $y = f(x) $ | 1 |
| (ii) | $y = [f(x)]^2$ | 1 |
| (iii) | $y = \frac{1}{f(x)}$ | 2 |
| (iv) | $y = \log_e f(x)$ | 2 |

- (b) The horizontal base of a solid is the area enclosed by the curve $|x|^{\frac{1}{2}} + |y|^{\frac{1}{2}} = 1$.
Vertical cross sections taken perpendicular to the x -axis are equilateral triangles with one side in the base.



- (i) Show that the volume of the solid is given by $V = 2\sqrt{3} \int_0^1 (1 - \sqrt{x})^4 dx$ 2
- (ii) Use the substitution of $u = 1 - \sqrt{x}$ to evaluate this integral. 3
- (c) The tangent AE is parallel to the chord DC .
- (i) Prove that $(AB)^2 = BC \cdot BE$ 3
- (ii) Hence or otherwise prove that $\frac{AC}{AE} = \sqrt{\frac{BC}{BE}}$ 1



Question 14 (15 marks) (Use a new page to write your answers)

- (a) The equation of an ellipse is given by $4x^2 + 9y^2 = 36$.
- (i) Find S and S' the foci of the ellipse 2
- (ii) Find the equations of the directrices M and M' 1
- (iii) Sketch the ellipse showing foci, directrices and axial intercepts. 2
- (iv) Let P be any point on the ellipse. 2
Show $SP + S'P = 6$
- (v) Find the equation of the chord of contact from an external point $(3, 2)$ 1
- (b) (i) Sketch the rectangular hyperbola $xy = c^2$, labelling the 1
point $P\left(ct, \frac{c}{t}\right)$ on it.
- (ii) Show that the equations of the tangent and normal to the hyperbola 3
at P are $x + t^2y = 2ct$ and $ty + ct^4 = t^3x + c$ respectively.
- (iii) If the tangent at P meets the coordinate axes at X and Y respectively 3
and the normal at P meets the lines $y = x$ and $y = -x$ at R and S respectively,
prove that the quadrilateral $RYSX$ is a rhombus.

Question 15 (15 marks) (Use a new page to write your answers)

- (a) When a certain polynomial is divided by $x+1$, $x-3$ the respective remainders are 6 and -2 . Find the remainder when this polynomial is divided by x^2-2x-3 . 3

- (b) The cubic equation $x^3+px+q=0$ has 3 non-zero roots α, β, γ . 3

Find, in terms of the constants p, q the values of

(i) $\alpha^2 + \beta^2 + \gamma^2$

(ii) $\alpha^3 + \beta^3 + \gamma^3$.

- (c) If α, β, γ are the roots of the equation $3x^3-5x^2-4x+3=0$, find the cubic equation with roots $\alpha-1, \beta-1, \gamma-1$. 3

- (d) A polynomial of degree n is given by $P(x) = x^n + ax - b$. It is given that the polynomial has a double root at $x = \alpha$.

- (i) Find the derived polynomial $P'(x)$ and show that $\alpha^{n-1} = -\frac{a}{n}$. 3

- (ii) Show that $\left(\frac{a}{n}\right)^n + \left(\frac{b}{n-1}\right)^{n-1} = 0$. 2

- (iii) Hence deduce that the double root is $\frac{bn}{a(n-1)}$. 1

Question 16 (15 marks) (Use a new page to write your answers)

- (a) For $a > 0$, $b > 0$, $c > 0$ and $d > 0$ and given that $\frac{a+b}{2} \geq \sqrt{ab}$, show that 2

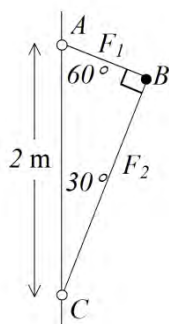
$$\frac{a+b+c+d}{4} \geq \sqrt[4]{abcd}$$

- (b) (i) Use De Moivre's theorem to express $\tan 5\theta$ in terms of powers of $\tan \theta$. 3

- (ii) Hence show that $x^4 - 10x^2 + 5 = 0$ has roots $\pm \tan \frac{\pi}{5}$ and $\pm \tan \frac{2\pi}{5}$. 2

- (iii) Deduce that $\tan \frac{\pi}{5} \cdot \tan \frac{2\pi}{5} \cdot \tan \frac{3\pi}{5} \cdot \tan \frac{4\pi}{5} = 5$ 1

- (c) A mass 10 kg, centre B is connected by light rods to sleeves A and C which revolve freely about the vertical axis AC but do not move vertically.



- (i) Given $AC = 2$ metres, show that the radius of the circular path of rotation of B is $\frac{\sqrt{3}}{2}$ metres. 1
- (ii) Find the tensions in the rods AB, BC when the mass makes 90 revolutions per minute about the vertical axis. 3
- (d) Given that $a_n = \sqrt{2 + a_{n-1}}$ for integers $n \geq 1$ and $a_0 = 1$, by mathematical induction prove that for $n \geq 1$:

$$\sqrt{2} < a_n < 2$$

Section 1

1) B

$$\begin{aligned} 2) \quad iz &= i(4+i) \\ &= 4i + i^2 \\ &= 4i - 1 \\ &= -1 + 4i \end{aligned}$$

$$\begin{aligned} \bar{iz} &= -1 - 4i \\ &= A \end{aligned}$$

3) A

$$\begin{aligned} 4) \quad \text{Let } y &= 0 \\ \frac{x^2}{9} &= 1 \\ x^2 &= 9 \\ x &= \pm 3 \end{aligned}$$

$$\begin{aligned} (\pm 3, 0) \\ A \end{aligned}$$

5) Normal and P(cp, $\frac{c}{p}$)

$$\therefore y - \frac{c}{p} = p^2(x - cp)$$

$$py - c = p^3(x - cp)$$

$$py - c = p^3x - p^4c$$

$$p^3x - py + c - cp^4 = 0$$

$$\begin{aligned} x &= ct \quad \frac{dx}{dt} = c \\ y &= \frac{c}{t} \quad \frac{dy}{dt} = -\frac{c}{t^2} \\ \therefore \frac{dy}{dx} &= -\frac{1}{t} \end{aligned}$$

$\therefore$ gradient of normal is t^2
in this case is p^2

B

$$6) \quad t = \tan \frac{x}{2}$$

$$\begin{aligned} \frac{dt}{dx} &= \frac{1}{2} \sec^2 \frac{x}{2} \\ &= \frac{1}{2} (1+t^2) \end{aligned}$$

$$dx = \frac{2dt}{1+t^2}$$

$$\begin{aligned} \int \sec x &= \int \frac{1+t^2}{1-t^2} \cdot \frac{2}{1+t^2} dt \\ &= \int \frac{2}{1-t^2} dt \\ &= \int \frac{A}{1-t} + \frac{B}{1+t} dt \end{aligned}$$

$$A(1+t) + B(1-t) = 2$$

$$\text{Let } t = -1 \quad \therefore B = 1$$

$$t = 1 \quad \therefore A = 1$$

$$\therefore \int \frac{1}{1-t} + \int \frac{1}{1+t} = -\ln(1-t) + \ln(1+t) = \ln \left| \frac{1+t}{1-t} \right| + C \quad B$$

$$7) \quad I_n = \int_0^{\frac{\pi}{2}} \sin^n x \, dx \quad 0 \leq x \leq \frac{\pi}{2}$$

$$\int_0^{\frac{\pi}{2}} \sin^{n-1} x \sin x \, dx$$

$$\begin{aligned} u &= \sin^{n-1} x & u' &= (n-1) \sin^{n-2} x \cos x \\ v' &= \sin x & v &= -\cos x \end{aligned}$$

$$I_n = [-\cos x \sin^{n-1} x] + (n-1) \int \cos^2 x \sin^{n-2} x \, dx$$

$$I_n = (n-1) \int (1 - \sin^2 x) \sin^{n-2} x \, dx$$

$$= (n-1) \int \sin^{n-2} x \, dx - (n-1) \int \sin^n x \, dx$$

$$I_n (1+n-1) = (n-1) \int \sin^{n-2} x \, dx$$

$$n I_n = n-1 I_{n-2}$$

$$I_n = \frac{n-1}{n} (I_{n-2})$$

A

8)



$$\int_0^5 \pi x y \, dx$$

$$= \int_0^5 \pi x x^3 \, dx$$

$$= \int_0^5 \pi x^4 \, dx$$

$$= \pi \left[\frac{x^5}{5} \right]_0^5$$

$$= \frac{32\pi}{5}$$

B

$$9) \quad \tan \theta = \frac{v^2}{rg}$$

$$= \frac{25^2}{100 \times 9.8}$$

$$= 0.6377$$

$$\theta = 32.32^\circ$$

B

10)

A $\alpha \beta \gamma$ satisfy $x^3 + 4x^2 - 2x - 5 = 0$ $\alpha^2 \beta^2 \gamma^2$ satisfy $(x^{\frac{1}{2}})^3 + 4(x^{\frac{1}{2}})^2 - 2(x^{\frac{1}{2}}) - 5 = 0$

$$x^{\frac{3}{2}} + 4x - 2x^{\frac{1}{2}} - 5 = 0$$

$$x^{\frac{3}{2}} - 2x^{\frac{1}{2}} = -4x + 5$$

$$x^{\frac{1}{2}}(x-2) = -4x + 5$$

$$x(x-2)^2 = (-4x+5)^2$$

$$x(x^2 - 4x + 4) = 16x^2 - 40x + 25$$

$$x^3 - 4x^2 + 4x = 16x^2 - 40x + 25$$

$$x^3 - 20x^2 + 44x - 25 = 0$$

B

Section 2. Question 11

a) (i) $\int \frac{t^2-1}{t^3} dt$

$$\int \frac{t^2}{t^3} - \int \frac{1}{t^3}$$

$$\int \frac{1}{t} - \int t^{-3}$$

$$\ln t - \frac{t^{-2}}{-2}$$

$$\ln t + \frac{1}{2t^2} + C$$

(ii) $\int \frac{dx}{\sqrt{6-x-x^2}}$

$$\int \frac{dx}{\sqrt{-(x^2+x-6)}}$$

$$\int \frac{dx}{\sqrt{-(x+\frac{1}{2})^2 - \frac{25}{4}}}$$

$$= \int \frac{dx}{\sqrt{\frac{25}{4} - (x+\frac{1}{2})^2}}$$

$$= \sin^{-1} \frac{x+\frac{1}{2}}{\frac{5}{2}}$$

$$= \sin^{-1} \left(\frac{2x+1}{5} \right) + C$$

b) i $\int_0^1 \frac{x}{(x+1)(2x+1)} dx$

$$\frac{A}{x+1} + \frac{B}{2x+1} = x$$

$$A(2x+1) + B(x+1) = x$$

$$\text{Let } x = -1 \quad -A = -1$$

$$A = 1$$

$$x = -\frac{1}{2} \quad \frac{1}{2}B = -\frac{1}{2}$$

$$B = -1$$

$$\int_0^1 \frac{1}{x+1} + \int_0^1 \frac{-1}{2x+1}$$

$$= \int_0^1 \frac{1}{x+1} - \frac{1}{2} \int_0^1 \frac{2}{2x+1} \left[\ln(x+1) - \frac{1}{2} \ln(2x+1) \right]_0^1$$

$$= \ln \left(\frac{2}{\sqrt{3}} \right) - \ln \frac{2\sqrt{3}}{5}$$

$$(ii) \int_0^{\frac{\pi}{4}} x \tan x \, dx \quad u=x \quad \frac{du}{dx}=1$$

$$\int_0^{\frac{\pi}{4}} x (\sec^2 x - 1) \, dx \quad \frac{dv}{dx} = \sec^2 x - 1 \quad v = \tan x - x$$

$$\therefore I = [x(\tan x - x)]_0^{\frac{\pi}{4}} - \int_0^{\frac{\pi}{4}} (\tan x - x) \, dx$$

$$= [x(\tan x - x)]_0^{\frac{\pi}{4}} - \int_0^{\frac{\pi}{4}} \frac{\sin x}{\cos x} - x \, dx$$

$$= [x(\tan x - x) + \ln(\cos x) + \frac{x^2}{2}]_0^{\frac{\pi}{4}}$$

$$= [x \tan x - \frac{x^2}{2} + \ln(\cos x)]_0^{\frac{\pi}{4}}$$

$$= \frac{\pi}{4} - \frac{\pi^2}{32} + \ln \frac{1}{\sqrt{2}}$$

$$C \quad I_n = \int_0^{\frac{\pi}{2}} x^n \cos x \, dx \quad u=x^n \quad \frac{du}{dx} = n x^{n-1}$$

$$\frac{dv}{dx} = \cos x \quad v = \sin x$$

$$[x^n \sin x]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} \sin x \cdot n x^{n-1} \, dx$$

$$\left(\frac{\pi}{2}\right)^n + n \int_0^{\frac{\pi}{2}} x^{n-1} \sin x \, dx \quad u=x^{n-1} \quad \frac{du}{dx} = (n-1)x^{n-2}$$

$$\left(\frac{\pi}{2}\right)^n + n \left[x \cos x \right]_0^{\frac{\pi}{2}} - n \int_0^{\frac{\pi}{2}} \cos x (n-1) x^{n-2} \, dx$$

$$= \left(\frac{\pi}{2}\right)^n + 0 - n(n-1) I_{n-2}$$

$$I_n = \left(\frac{\pi}{2}\right)^n - n(n-1) I_{n-2}$$

$$(i) \quad A = I_4$$

$$= \left(\frac{\pi}{2}\right)^4 - 4 \times 3 \left[\left(\frac{\pi}{2}\right)^2 - 2 \int_0^{\frac{\pi}{2}} \cos x \, dx \right]$$

$$= \left(\frac{\pi}{2}\right)^4 - 3\pi^2 + 24 \left[\sin x \right]_0^{\frac{\pi}{2}}$$

$$= \left(\frac{\pi}{2}\right)^4 - 3\pi^2 + 24$$

Question 12

$$a(i) \quad -\sqrt{2} (\sqrt{2} - \sqrt{2}i)^2$$

$$= -\sqrt{2} (2 - 4i + 2i^2)$$

$$= -\sqrt{2} (0 - 4i)$$

$$= 4\sqrt{2}i$$

$$(i) \quad \begin{array}{c} \sqrt{2} \\ \cos \frac{\pi}{4} \\ 2 \end{array} \quad \begin{array}{c} \sqrt{2} \\ \sin \frac{\pi}{4} \end{array}$$

$$= -\frac{\pi}{4}$$

$$(ii) \quad \frac{z}{z+w}$$

$$\frac{\sqrt{2} - \sqrt{2}i}{\sqrt{2} - \sqrt{2}i + \sqrt{2}}$$

$$= \frac{\sqrt{2} - \sqrt{2}i}{-\sqrt{2}i} \times \frac{\sqrt{2}i}{\sqrt{2}i}$$

$$= \frac{2i - 2i^2}{-2i^2}$$

$$= \frac{2i + 2}{2}$$

$$= 1 + i$$

$$(iv) \quad |z|$$

$$= \sqrt{\sqrt{2}^2 + \sqrt{2}^2}$$

$$= \sqrt{4}$$

$$= 2$$

$$(v) \quad z^{10} = 2 \left(\cos -\frac{\pi}{4} \right)^{10}$$

$$= 2^{10} \cos -\frac{10\pi}{4} = 1024 \cos -\frac{2\pi}{4} = \cos -\frac{\pi}{2} = -i 1024$$

$$b) (a+ib)^2 = 5-12i$$

$$a^2+2abi-b^2=5-12i$$

$$a^2-b^2=5 \quad - (1)$$

$$2ab=-12 \quad - (2)$$

$$ab=-6$$

$$b = -\frac{6}{a}$$

$$a^2 - \frac{36}{a^2} = 5$$

$$a^4 - 5a^2 - 36 = 0$$

$$(a^2-9)(a^2+4)=0$$

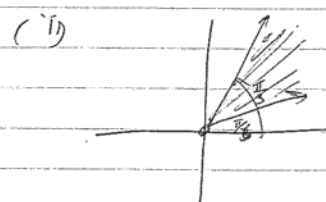
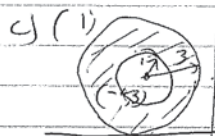
$$a^2=9 \quad \therefore a=\pm 3$$

$$2(\pm 3)b = -12$$

$$\therefore b = (\pm 2)$$

$$a=3 \quad b=2$$

$$\text{or } a=-3 \quad b=2$$



$$d) i) \frac{1+\cos\theta+i\sin\theta}{1-\cos\theta-i\sin\theta} = \frac{2\cos^2\frac{\theta}{2}+i2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{2\sin^2\frac{\theta}{2}-i2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}$$

$$= \frac{2\cos\frac{\theta}{2}(\cos\frac{\theta}{2}+i\sin\frac{\theta}{2})}{2\sin\frac{\theta}{2}(\sin\frac{\theta}{2}-i\cos\frac{\theta}{2})}$$

$$= \frac{\cos\frac{\theta}{2}(\cos\frac{\theta}{2}+i\sin\frac{\theta}{2})}{-\sin\frac{\theta}{2}(\cos\frac{\theta}{2}+i\sin\frac{\theta}{2})}$$

$$\text{since } \sin x - i\cos x = -i(\cos x + i\sin x)$$

$$= i\cot\frac{\theta}{2} \quad \text{since } \frac{1}{-i} = i$$

$$\left(\frac{z-1}{z+1}\right)^8 = -1 \Rightarrow \frac{z-1}{z+1} = \sqrt[8]{-1}$$

$$\frac{z-1}{z+1} = \sqrt[8]{\text{cis}(\pi+2k\pi)} = \text{cis}\frac{(2k+1)\pi}{8} \quad k=0, \pm 1, \pm 2$$

$$z-1 = \left(\text{cis}\frac{(2k+1)\pi}{8}\right)(z+1)$$

$$z-1 = \text{cis}\frac{(2k+1)\pi}{8}z + \text{cis}\frac{(2k+1)\pi}{8}$$

$$z\left(1 - \text{cis}\frac{(2k+1)\pi}{8}\right) = \text{cis}\frac{(2k+1)\pi}{8} + 1$$

$$\therefore z = \frac{1 + \text{cis}\frac{(2k+1)\pi}{8}}{1 - \text{cis}\frac{(2k+1)\pi}{8}}$$

$$= i\cot\frac{(2k+1)\pi}{16} \quad \text{from (i)}$$

$$= i\cot\frac{\pi}{16}, i\cot\frac{3\pi}{16}$$

$$= \pm i\cot\frac{\pi}{16}, \pm i\cot\frac{3\pi}{16} \quad \text{since } \cot x \text{ is an odd function}$$

d) i)

Alternatively Let $t = \tan\frac{\theta}{2}$.

$$\text{L.H.S } \frac{1 + \frac{1-t^2}{1+t^2} + i\frac{2t}{1+t^2}}{1 - \frac{1-t^2}{1+t^2} - i\frac{2t}{1+t^2}}$$

$$= \frac{2+i2t}{2t^2-i2t}$$

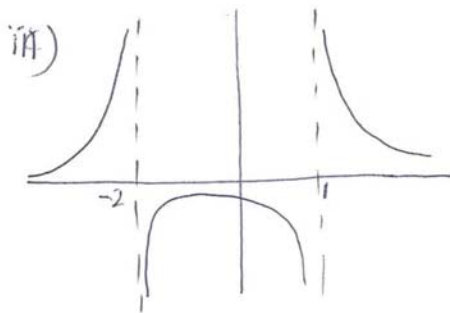
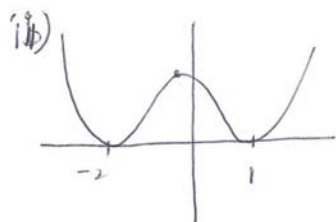
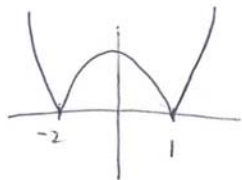
$$= \frac{1+it}{t^2-it}$$

$$= \frac{i(t-i)}{t(t-i)}$$

$$= \frac{i}{t} = i\cot\frac{\theta}{2} = \text{R.H.S}$$

Question 13

i) $y = |f(x)|$



iv) $y = \log f(x)$

As $x \rightarrow -2$ or 1 $f(x) \rightarrow 0$ $\log f(x) \rightarrow +\infty$

crosses x axis when $\ln f(x) = 0$

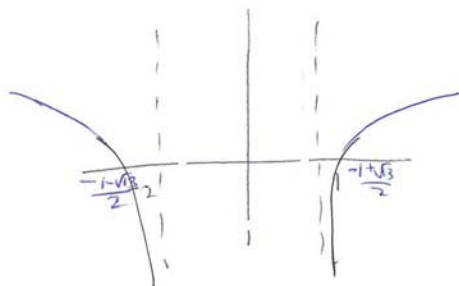
ie when

$$f(x) = 1$$

$$x^2 + x - 2 = 1$$

$$x^2 + x - 3 = 0$$

$$x = \frac{-1 \pm \sqrt{13}}{2}$$



(b) (i)



$$A = 2y \cdot \sqrt{3}y \times \frac{1}{2}$$

$$A = \sqrt{3}y^2$$

$$\int V = \int \sqrt{3}y^2 dx$$

$$V = \int_{x=0}^1 \sqrt{3}y^2 dx$$

$$= \int_0^1 \sqrt{3}y^2 dx$$

$$= \int_0^1 \sqrt{3}y^2 dx$$

$$= \sqrt{3} \int_0^1 (1-\sqrt{x})^4 dx$$

double since both sides of y axis

$$= 2\sqrt{3} \int_0^1 (1-\sqrt{x})^4 dx$$

(ii) $u = 1 - \sqrt{x}$ $x = (1-u)^2$ $x=0 \Rightarrow u=1$
 $dx = -2(1-u)du$ $x=1 \Rightarrow u=0$

$$= 2\sqrt{3} \int_1^0 u^4 \times -2(1-u) du$$

$$= 4\sqrt{3} \int_0^1 u^4 - u^5 du$$

$$= 4\sqrt{3} \left[\frac{1}{5} u^5 - \frac{1}{6} u^6 \right]_0^1$$

$$= 4\sqrt{3} \left[\frac{1}{5} - \frac{1}{6} \right]$$

$$= \frac{4\sqrt{3}}{2 \times 30} = \frac{2\sqrt{3}}{15}$$

C (i) Aim: prove $(AB)^2 = BC \cdot BE$

prod: In $\triangle ABC$ and $\triangle EBA$

$\angle AEB = \angle CDE$ (alternate $\angle$'s on parallel lines)

$\angle CDE = \angle CAB$ (angles in the same segment)

$\therefore \angle AEB = \angle CAB$ — A.

$\angle BAE = \angle BCA$ (angle in the alternate segment)

$\therefore \triangle ABC \sim \triangle EBA$ equiangular.

$$\therefore \frac{AB}{BC} = \frac{BE}{AB} \quad \text{or } AB^2 = BC \cdot BE$$

(ii) $\frac{AC}{AE} = \frac{BC}{BA}$ since similar triangles have sides in proportion.

$$AB^2 = BC \cdot BE$$

$$\therefore AB = \sqrt{BC \cdot BE}$$

$$\frac{AC}{AE} = \frac{BC}{\sqrt{BC \cdot BE}}$$

$$= \frac{BC}{\sqrt{BC} \sqrt{BE}}$$

$$\frac{AC}{AE} = \frac{\sqrt{BC}}{\sqrt{BE}}$$

Question 14

(a) $4x^2 + 9y^2 = 36$

(i) $\frac{x^2}{9} + \frac{y^2}{4} = 1$

$$a^2 = 9 \quad b^2 = 4 \quad b^2 = a^2(1 - e^2)$$

$$4 = 9(1 - e^2)$$

$$\frac{4}{9} = 1 - e^2$$

$$e^2 = \frac{5}{9}$$

$$e = \frac{\sqrt{5}}{3}$$

$$S(ae, 0) \quad S'(ae, 0)$$

$$S(\sqrt{5}, 0) \quad S'(-\sqrt{5}, 0)$$

(ii) $x = \pm \frac{a}{e}$

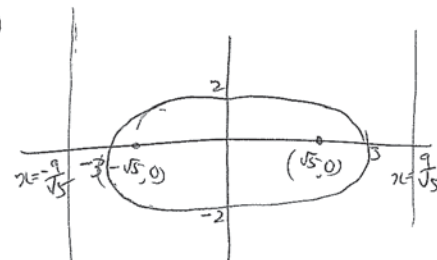
$$= \pm \frac{3}{\frac{\sqrt{5}}{3}}$$

$$= \pm \frac{9}{\sqrt{5}}$$

$$M: x = \frac{9}{\sqrt{5}}$$

$$M': x = -\frac{9}{\sqrt{5}}$$

(iii)



$$(IV) \quad SP + SP' = 6$$

$$PS = e PM$$

$$P'S' = e P'M' \quad \text{where } M \text{ and } M' \text{ are the feet}$$

of the perpendiculars from P to m and m' .

$$PS + P'S' = e(PM + P'M')$$

$$= e(MM')$$

$$= e\left(\frac{a}{e} + \frac{a}{e}\right)$$

$$= \frac{2ae}{e}$$

$$PS + P'S' = 2a$$

$$a = 3.$$

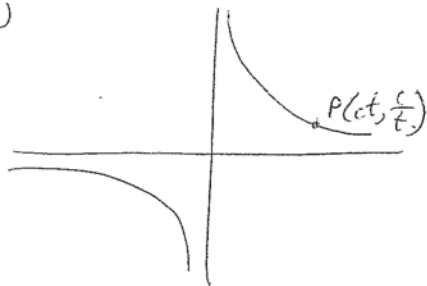
$$\therefore SP + SP' = 6$$

$$(V) \quad \frac{x x_0}{a^2} + \frac{y y_0}{b^2} = 1$$

$$\frac{3x}{9} + \frac{2y}{4} = 1$$

$$\frac{x}{3} + \frac{y}{2} = 1.$$

b(i)



$$(Vi) \quad xy = c^2$$

$$y = \frac{c^2}{x}$$

$$\frac{dy}{dx} = -\frac{c^2}{x^2}$$

$$\text{at } P, \text{ grad tangent} = -\frac{1}{t^2}$$

$$\therefore \text{grad normal} = t^2.$$

$$\text{Eqn of tangent } y - \frac{c}{t} = -\frac{1}{t^2}(x - ct)$$

$$t^2 y - ct = -x + ct$$

$$\text{Equation of normal } \frac{x + t^2 y}{1} = 2ct$$

$$y - \frac{c}{t} = t^2(x - ct)$$

$$ty - c = t^3 x - ct^4$$

$$ty + ct^4 = t^3 x + c.$$

$$(iii) \quad X(2ct, 0)$$

$$Y(0, \frac{2c}{t})$$

$$R\left(\frac{c(t^2+1)}{t}; \frac{c(t^2-1)}{t}\right)$$

$$S\left(\frac{c(t^2-1)}{t}, -\frac{c(t^2-1)}{t}\right)$$

$$\text{Midpoint } XY = (ct, \frac{c}{t})$$

$$\text{Midpoint } RS = \left(\frac{2ct^2}{t}, \frac{2c}{t}\right)$$

$$= (ct, \frac{c}{t})$$

$$\text{Grad } XY = \frac{\frac{2c}{t}}{-2ct} = -\frac{1}{t^2}$$

$$\text{Grad } RS = \frac{\frac{2ct^2}{t}}{\frac{2c}{t}} = t^2$$

$\therefore RS \perp XY$
 $\therefore RPSX$ is a rhombus

Question 15

a) $P(x) = (x+1)(x-3) \cdot Q(x) + ax+b$

$$P(-1) = -a+b = 6 \quad \text{--- (1)}$$

$$P(3) = 3a+b = -2 \quad \text{--- (2)}$$

$$\textcircled{1} - \textcircled{2} \quad -4a = 8$$

$$a = -2$$

$$\therefore b = 4$$

$$x^2 - 2x - 3 = (x+1)(x-3)$$

when divided by $x^2 - 2x - 3$

$$\therefore R(x) = -2x + 4$$

b(i) $\alpha, \beta, \gamma \quad \dots \quad \alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha)$

$$= 0 - 2p$$

$$\alpha^3 + \beta^3 + \gamma^3 = -2p$$

(ii)

$$\alpha^3 + p\alpha + q = 0$$

$$\beta^3 + p\beta + q = 0$$

$$\gamma^3 + p\gamma + q = 0$$

$$\alpha^3 + \beta^3 + \gamma^3 + p(\alpha + \beta + \gamma) + 3q = 0$$

$$\alpha^3 + \beta^3 + \gamma^3 + p \cdot 0 + 3q = 0$$

$$\therefore \alpha^3 + \beta^3 + \gamma^3 = -3q$$

(c) Let $y = x-1$

$$\therefore x = y+1$$

$$3(y+1)^3 - 5(y+1)^2 - 4(y+1) + 3 = 0$$

$$3(y^3 + 3y^2 + 3y + 1) - 5(y^2 + 2y + 1) - 4(y+1) + 3 = 0$$

$$3y^3 + 9y^2 + 9y + 3 - 5y^2 - 10y - 5 - 4y - 4 + 3 = 0$$

$$3y^3 + 4y^2 - 5y - 3 = 0$$

in terms of x

$$3x^3 + 4x^2 - 5x - 3 = 0$$

(d)(i) $P(x) = x^n + ax - b$

double root if $x = \alpha$

$$P(\alpha) = n\alpha^{n-1} + a$$

$$P'(\alpha) = n(n-1)\alpha^{n-2}$$

$$\text{note } P(\alpha) = 0 \Rightarrow \alpha^n + a\alpha - b = 0$$

$$P'(\alpha) = 0 \Rightarrow n\alpha^{n-1} + a = 0$$

$$\therefore \alpha^{n-1} = -\frac{a}{n}$$

(ii) $P(\alpha) = \alpha^n + a\alpha - b = 0 \quad \text{--- (1)}$

$$P'(\alpha) = n\alpha^{n-1} + a = 0 \quad \text{--- (2)}$$

$$n\alpha^n + a\alpha = 0 \quad \text{--- (2)}$$

$$\textcircled{1} - \textcircled{2} \quad (1-n)\alpha^n - b = 0$$

$$\alpha^n = \frac{b}{1-n} \quad \text{--- (3)}$$

$$\text{also } \alpha^{n-1} = -\frac{a}{n} \quad \text{--- (4)}$$

from (3) $(\alpha^n)^{n-1} = \left(\frac{b}{1-n}\right)^{n-1}$

from (4) $(\alpha^{n-1})^n = \left(-\frac{a}{n}\right)^n$

$$\left(\frac{b}{1-n}\right)^{n-1} = \left(-\frac{a}{n}\right)^n$$

$$\left(-\frac{b}{n-1}\right)^{n-1} = (-1)^n \left(\frac{a}{n}\right)^n$$

$$(-1)^{n-1} \left(\frac{b}{n-1}\right)^{n-1} = (-1)^{n-1} \left(\frac{a}{n}\right)^n$$

$$\left(\frac{a}{n}\right)^n + \left(\frac{b}{n-1}\right)^{n-1} = 0$$

(iii) double root is α

$$\alpha = \frac{a^n}{\alpha^{n-1}} \leftarrow \text{use (3)}$$

$$\alpha = \frac{\alpha^n}{\alpha^{n-1}} \leftarrow \text{use (4)}$$

$$= \left(\frac{b}{1-n}\right) / -\frac{a}{n}$$

$$\alpha = \frac{bn}{-a(1-n)} = \frac{bn}{a(n-1)}$$

Question 16.

(a) Let $x = \frac{a+b}{2}$

$$y = \frac{c+d}{2}$$

$$\therefore \frac{x+y}{2} = \frac{a+b+c+d}{4}$$

$$\text{Now } \frac{x+y}{2} \geq \sqrt{xy}$$

$$\therefore \frac{a+b+c+d}{4} \geq \sqrt{\sqrt{ac}\sqrt{bd}}$$

$$\frac{a+b+c+d}{4} \geq \sqrt[4]{abcd}$$

b) (i) $\text{cis } 5\theta = (\text{cis } \theta)^5$

$$\cos 5\theta + i \sin 5\theta = \cos^5 \theta + i 5 \cos^4 \theta \sin \theta - 10 \cos^3 \theta \sin^2 \theta - i 10 \cos^2 \theta \sin^3 \theta + 5 \cos \theta \sin^4 \theta + i \sin^5 \theta$$

$$\therefore \cos 5\theta = \cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta$$

$$\sin 5\theta = 5 \cos^4 \theta \sin \theta - 10 \cos^2 \theta \sin^3 \theta + \sin^5 \theta$$

$$\tan 5\theta = \frac{5 \cos^4 \theta \sin \theta - 10 \cos^2 \theta \sin^3 \theta + \sin^5 \theta}{\cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta}$$

$$= \frac{5 \tan \theta - 10 \tan^3 \theta + \tan^5 \theta}{1 - 10 \tan^2 \theta + 5 \tan^4 \theta}$$

$$= \tan \theta \cdot \frac{5 - 10 \tan^2 \theta + \tan^4 \theta}{1 - 10 \tan^2 \theta + 5 \tan^4 \theta}$$

(ii) Let $x = \tan \theta$ then $\tan 5\theta = 0 \therefore x^4 - 10x^2 + 5 = 0$

$$5\theta = 0 \text{ or } \pi \text{ or } 2\pi, \dots$$

$$\theta = \frac{\pi}{5}, \frac{2\pi}{5}, \frac{3\pi}{5}, \frac{4\pi}{5}$$

$$\tan \frac{3\pi}{5} = -\tan\left(\pi - \frac{3\pi}{5}\right)$$

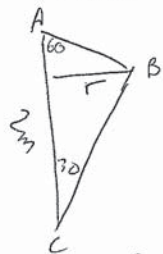
$$= -\tan \frac{2\pi}{5} \propto \tan \frac{4\pi}{5} = -\tan \frac{\pi}{5}$$

$$\text{root } x = \pm \tan \frac{\pi}{5} \text{ and } \pm \tan \frac{2\pi}{5}$$

product of roots $\left(\tan \frac{\pi}{5}\right)\left(\tan \frac{2\pi}{5}\right)\left(\tan \frac{3\pi}{5}\right)\left(\tan \frac{4\pi}{5}\right) = 5 = 5.$

$$\text{also } f = \frac{e}{a}$$

(i) Question 16(c)



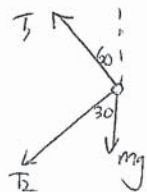
$$\cos 60 = \frac{AB}{2}$$

$$AB = 1$$

$$\sin 60 = \frac{r}{AB}$$

$$r = \frac{\sqrt{3}}{2}$$

(ii) Let tension in rods AB and BC be T_1 and T_2 respectively



$$\sum F_v = 0$$

$$T_1 \cos 60 = T_2 \cos 30 + mg$$

$$T_1 \left(\frac{1}{2}\right) = T_2 \left(\frac{\sqrt{3}}{2}\right) + 10g$$

$$T_1 = T_2 \sqrt{3} + 20g$$

$$\sum F_H = mr\omega^2$$

$$T_1 \sin 60 + T_2 \sin 30 = mr\omega^2$$

$$T_1 \left(\frac{\sqrt{3}}{2}\right) + T_2 \left(\frac{1}{2}\right) = 10 \left(\frac{\sqrt{3}}{2}\right) (3\pi)^2$$

$$T_1 = 90\pi^2 - \frac{T_2}{\sqrt{3}}$$

$$T_2 \sqrt{3} + 20g = 90\pi^2 - \frac{T_2}{\sqrt{3}}$$

$$T_2 = \frac{5\sqrt{3}}{2} (9\pi^2 - 2g)$$

$$T_1 = \frac{5}{2} (27\pi^2 + 2g)$$

so tensions in AB and BC are $\frac{5}{2} (27\pi^2 + 2g) \text{ N}$ and $\frac{5\sqrt{3}}{2} (9\pi^2 - 2g) \text{ N}$

16d). $n=1$

$$a_1 < \sqrt{2+a_0} = \sqrt{3}$$

since $\sqrt{2} < \sqrt{3} < 2$ is true for $n=1$.

Assume true for $n=k$

$$\sqrt{2} < a_k < 2 \quad (A)$$

Now prove true for $n=k+1$

$$\sqrt{2} < a_{k+1} < 2 \quad (B)$$

From (A) $\sqrt{2} < a_k < 2$

$$2 + \sqrt{2} < 2 + a_k < 4$$

$$\sqrt{2 + \sqrt{2}} < \sqrt{2 + a_k} < 2$$

$$\sqrt{2 + \sqrt{2}} < a_{k+1} < 2$$

$$\text{Now } 2 < 2 + \sqrt{2} \Rightarrow \sqrt{2} < \sqrt{2 + \sqrt{2}}$$

$$\therefore \sqrt{2} < a_{k+1} < 2$$

$\therefore$ proved by mathematical induction.